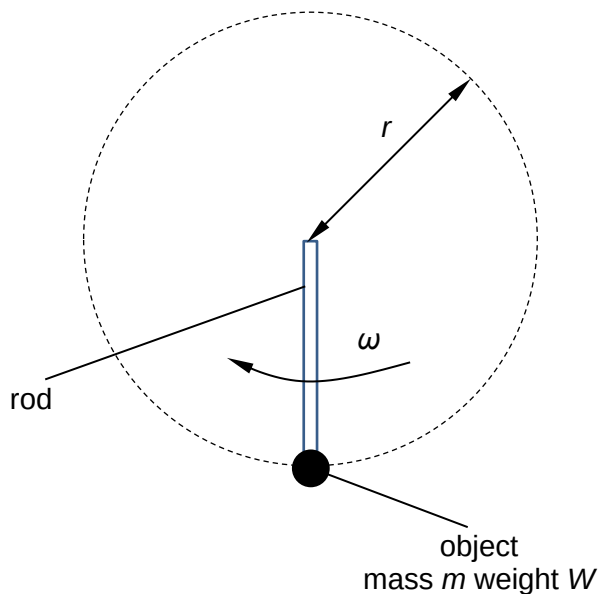


9

An object of mass m and weight W is fixed to one end of a light rod which rotates in a vertical circle of radius r . The object has an angular velocity when it is vertically below the centre of the circle



What is the tension in the rod acting upwards on the object when the object is vertically below the centre of the circle?

- | | | | | | | | |
|----------|------------------|----------|------------------|----------|------------------|----------|--------------|
| A | $W + mr\omega^2$ | B | $W - mr\omega^2$ | C | $mr\omega^2 - W$ | D | $mr\omega^2$ |
|----------|------------------|----------|------------------|----------|------------------|----------|--------------|

Answer: A

T – tension in the rod

At the bottom of the circular path, $(T - W)$ provides the centripetal force.

$$(T - W) = mr\omega^2 \quad \Rightarrow \quad T = W + mr\omega^2$$

[Turn over

10	Two isolated stars of mass M and $9M$ are separated in space by a distance d between the
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